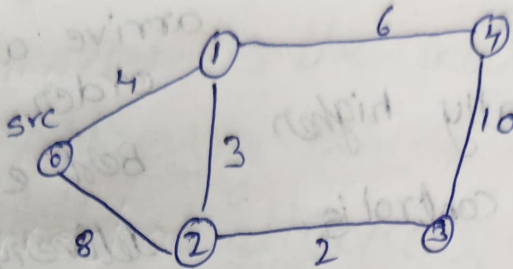


HA-3

24000325/3

1) Apply for Dijkstra's shortest path algorithm and explain every step to calculate the shortest path from the source node to all other nodes.



1) Start at Node 0 $\rightarrow$ Distance = 0

2) From 0 smallest Node is $1 \times 4 + 3 = 7$

3) Pick next node $\rightarrow$ node 2(7)

$$3 = 7 + 2 = 9$$

4) Next node is node 3(4) $\rightarrow$ 10

5) Next node 4(10)

Final shortest distance from 0

$$0 \rightarrow 0$$

$$0 \rightarrow 1 = 4$$

$$0 \rightarrow 2 = 7$$

$$0 \rightarrow 3 = 9$$

$$0 \rightarrow 4 = 10$$

2) Compare Distance vector routing with Link state routing in terms of convergence time and overhead
what is the count-to-infinity problem in distance vector routing? Explain with a suitable network example

A) Distance vector :-

- Routers share routing tables with neighbors
- Uses Bellman-Ford

Link state :-

- Routers share routing full topology
- Uses Dijkstra

Count to Infinity problem :-

- $A \rightarrow B \rightarrow C$ path exists
- Link between B and C breaks
- B tells A "C unreachable"
- A still thinks "C via B"
- Distance keep increasing $\rightarrow \infty$

3) Illustrate the DNS lookup process from scratch when the local DNS cache does not contain the requested domain?

A) when the local DNS cache is empty
User enters a domain

Browser/OS sends request to local DNS
Resolver queries root name server $\rightarrow$ gets TLD server info

Resolver queries Authoritative DNS server gets IP address

- IP is returned to browser, which leads to website

- Result is cached for future use

Flow:-

User $\rightarrow$ Resolver $\rightarrow$ Root $\rightarrow$ TLD $\rightarrow$ Authoritative
 $\rightarrow$ IP $\rightarrow$ Browser.

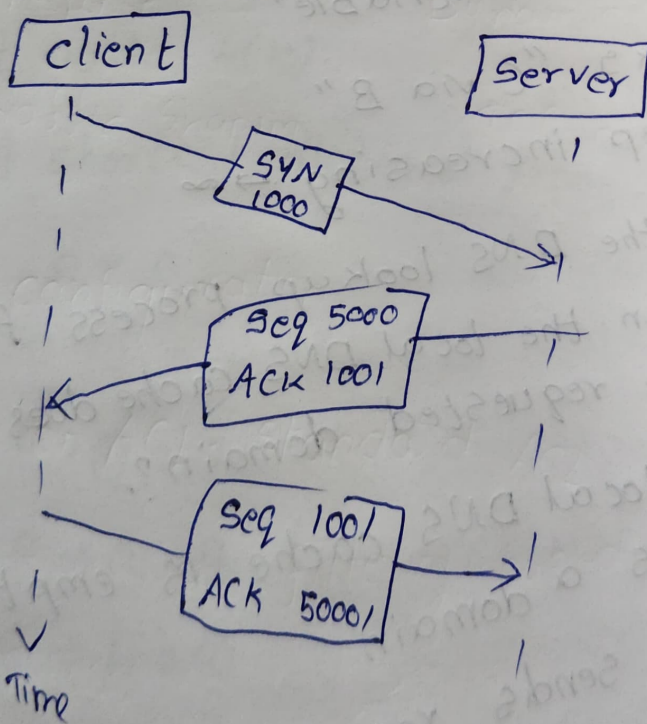
4) TCP connection

a) Three way handshake

1) client $\rightarrow$ server: SYN, seq=1000

2) server $\rightarrow$ client: SYN+ACK, Seq=5000
ACK=1001

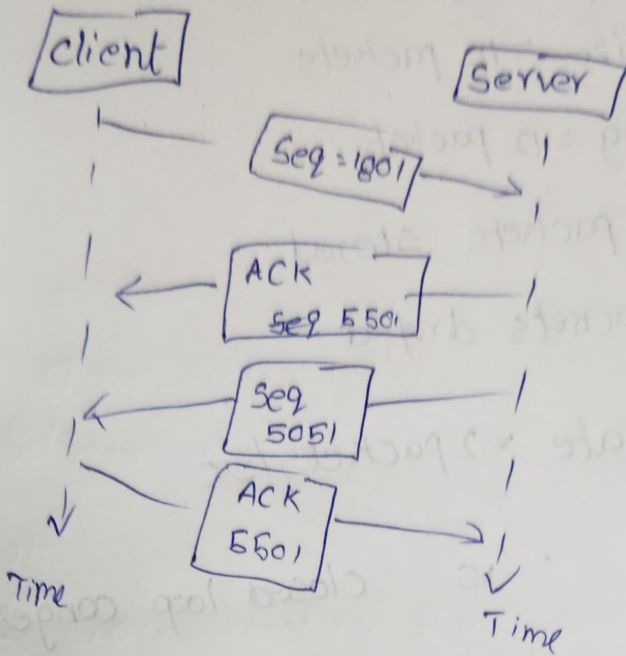
3) client $\rightarrow$ server: ACK, seq:1001, ACK=5000



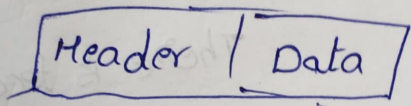
b) Data transfer

1) client

4) server ACK=1801, seq 5001 & client ACK=5501



5) TCP Header files



Source port address	Destination port address	→ identify
Sequence number		→ order of data
Acknowledgement number		→ confirms data
M L E N Res G v R A C K P S H R S T S Y N F I N window size		→ flow control
check sum	urgent pointer	
options & padding		→ error detection

6) Leaky Bucket algorithm

1) Bucket size = 10 packets

Incomming = 12 packets

Result: 10 packets stored

2 packets dropped

output rate = 2 packets/sec

7) open loop vs closed loop congestion control

Open loop

closed loop

No feedback

There is feedback

Simple complexity

Complex

fixed response

Dynamic response

Ex:- Traffic Shaping

TCP congestion control

8) ACL configuration

1) Types of ACL: standard ACL

Because we are filtering based only on Source IP address.

2) ACL configuration

```
Router(config)# access-list 10 deny 172.5.6.2  
Router(config)# access list 10 permit any.
```